```
import matplotlib.pyplot as plt
import numpy as np

def f(x):
    return(x+3)**2

def grad_f(x):
    return 2*(x+3)

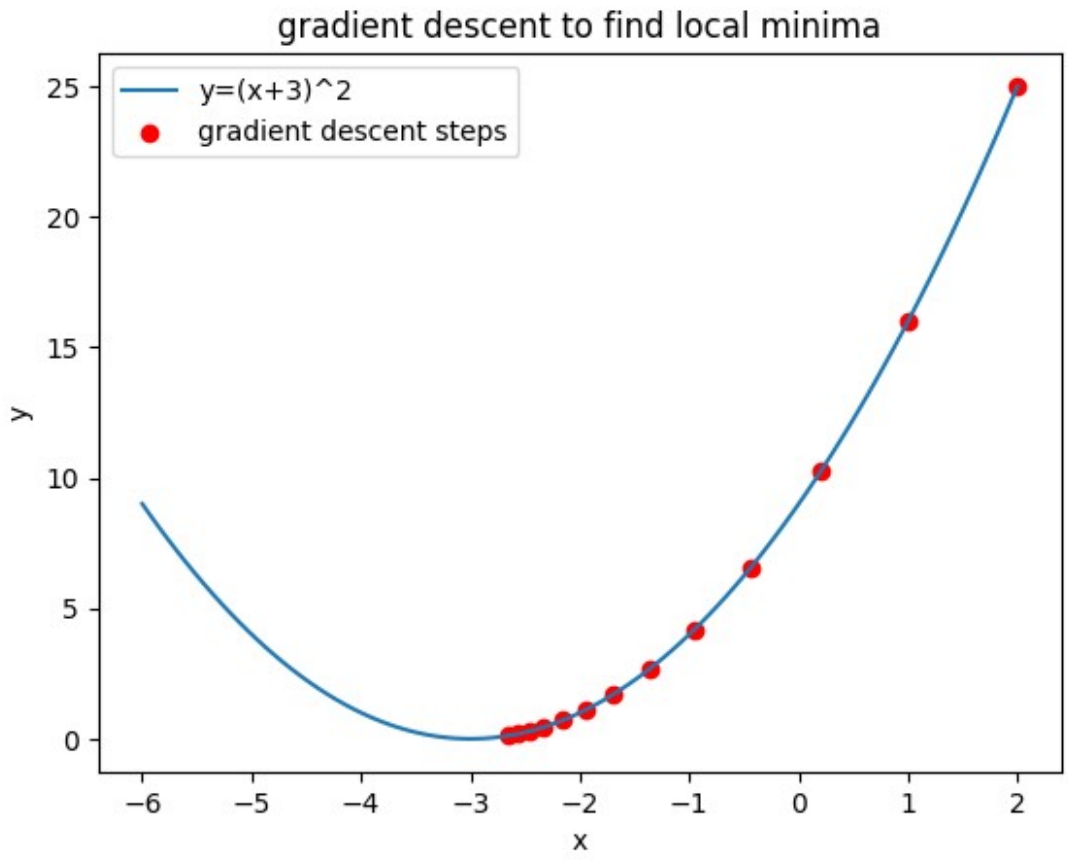
def gradient_descent(start_x=2, learning_rate=0.1,max_iter=50,tol=1e-
1):
        x=start_x
        x_history=[x]
        for i in range(max_iter):
            gradient=grad_f(x)
            x_new=x-learning_rate*gradient
            x_history.append(x_new)
            if abs(x_new-x)<tol:
                break
            x=x_new
        return x,f(x),x_history

min_x,min_y,x_steps=gradient_descent()
print("local minima at x=",min_x)
print("Minimum value y=",min_y)
```

```
local minima at x= -2.5705032704
Minimum value y= 0.1844674407370954
```

```
x_vals=np.linspace(-6,2,100)
y_vals=f(x_vals)

plt.plot(x_vals,y_vals,label='y=(x+3)^2')
plt.scatter(x_steps,[f(x) for x in
x_steps],color='red',label='gradient descent steps')
plt.title("gradient descent to find local minima")
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.show()
```


gradient descent to find local minima
y=(x+3)^2
gradient descent steps
25
20
15
10
5
0
y
−6
−5
−4
−3
−2
−1
0
1
2
x